

§7.3 - Trigonometric Substitution

Example #1

If

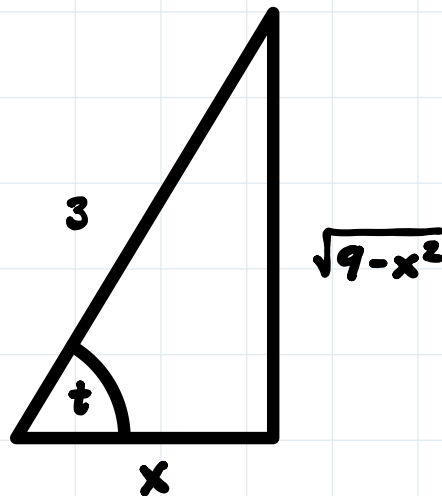
$$\tan(t) = \frac{\sqrt{9 - x^2}}{x}$$

for $x \in (0, 3)$ what is

$$\sin(t)$$

in terms of x ?

Take t to be the acute angle shown below. Then



$$\cos(t) = \frac{\text{adjacent}}{\text{hypotenuse}} \quad \text{and} \quad \sin(t) = \frac{\text{opposite}}{\text{hypotenuse}}$$

From our figure,

$$\sin(t) = \frac{\sqrt{9-x^2}}{3}, \quad x \in (0, 3)$$

$$\cos(t) = \frac{x}{3}, \quad x \in (0, 3)$$

and likewise

$$t = \arccos\left(\frac{x}{3}\right)$$

$$= \arctan\left(\frac{\sqrt{9-x^2}}{x}\right)$$

$$= \arcsin\left(\frac{\sqrt{9-x^2}}{3}\right).$$

Example #2 (Integral involving $a^2 - x^2$)

Find

$$\int \sqrt{1 - x^2} \, dx.$$

Our integrand looks like a semi circle and trig functions are a natural fit because the curve is easy to describe by a central angle.

Let

$x = \sin(t)$ $dx = \cos(t) dt$ for $t \in [-\pi/2, \pi/2]$
then

$$\begin{aligned}(1^2 - x^2)^{1/2} &= (1 - \sin^2(t))^{1/2} \\ &= (\cos^2(t))^{1/2} \\ &= |\cos(t)| \\ &= \cos(t)\end{aligned}$$

since

$$\cos^2(t) + \sin^2(t) = 1$$

and

$$\cos(t) \geq 0 \text{ for } t \in [-\pi/2, \pi/2].$$

Next

$$\int (1 - x^2)^{1/2} dx = \int \cos(t) \cos(t) dt = \int \cos^2(t) dt$$

and

$$\cos^2(t) = \frac{1}{2} (1 + \cos(2t))$$

so

$$\begin{aligned}\int (1-x^2)^{1/2} dx &= \frac{1}{2} \int 1 + \cos(2t) dt \\ &= \frac{1}{2}t + \frac{1}{4}\sin(2t) + C\end{aligned}$$

Now

$$x = \sin(t) \Rightarrow t = \arcsin(x)$$

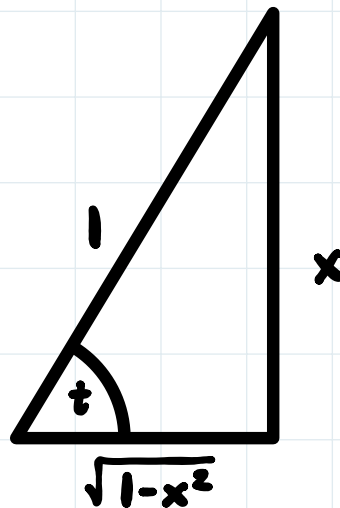
and

$$\sin(2t) = 2\cos(t)\sin(t)$$

yield

$$\int (1-x^2)^{1/2} dx = \frac{1}{2}t + \frac{1}{2}\cos(t)\sin(t) + C.$$

But what is $\cos(t)$ in terms of x ? Look at a reference triangle:



$$\Rightarrow \cos(t) = \sqrt{1-x^2}$$

so

$$\begin{aligned}\int (1-x^2)^{1/2} dx &= \frac{1}{2}t + \frac{1}{2}\cos(t)\sin(t) + C \\ &= \frac{1}{2}\arcsin(x) + \frac{1}{2}\sqrt{1-x^2} \cdot x + C\end{aligned}$$

Example #3 (Integral involving $a^2 - x^2$)

Find

$$\int \sqrt{36 - x^2} \, dx.$$

Let

$$x = 6 \sin(t) \quad dx = 6 \cos(t) dt$$

for $t \in [-\pi/2, \pi/2]$ and

$$\begin{aligned}(6^2 - x^2)^{1/2} &= (6^2 - (6 \sin(t))^2)^{1/2} \\&= 6(1 - \sin^2(t))^{1/2} \\&= 6(\cos^2(t))^{1/2} \\&= 6|\cos(t)| \\&= 6\cos(t)\end{aligned}$$

since $\cos^2(t) + \sin^2(t) = 1$ and $\cos(t) \geq 0$ over $[-\pi/2, \pi/2]$.

Then

$$\begin{aligned}\int (6^2 - x^2)^{1/2} dx &= \int 6\cos(t) \cdot 6\cos(t) dt \\&= \int 36\cos^2(t) dt \\&= 36 \cdot \frac{1}{2} \int 1 + \cos(2t) dt \\&= 18t + \frac{18}{2} \sin(2t) + C\end{aligned}$$

because

$$\cos^2(t) = \frac{1 + \cos(2t)}{2}.$$

Now

$$x = 6 \sin(t) \Rightarrow \frac{x}{6} = \sin(t) \Rightarrow t = \arcsin\left(\frac{x}{6}\right)$$

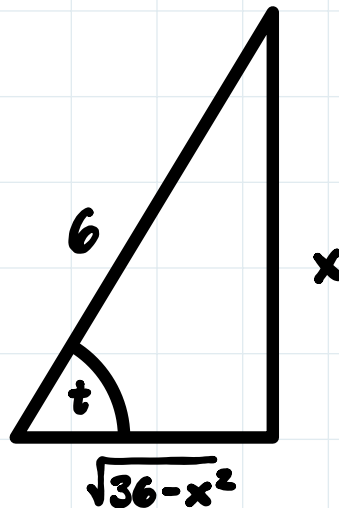
and

$$\sin(2t) = 2 \cos(t) \sin(t)$$

give

$$\int (6^2 - x^2)^{1/2} dx = 18t + 18 \cos(t) \sin(t) + C.$$

What is $\cos(t)$ in terms of x ?



$$\Rightarrow \cos(t) = \frac{\sqrt{36 - x^2}}{6}$$

We get

$$\begin{aligned} \int (6^2 - x^2)^{1/2} dx &= 18 \arcsin\left(\frac{x}{6}\right) + 18 \frac{\sqrt{36 - x^2}}{6} \left(\frac{x}{6}\right) + C \\ &= 18 \arcsin\left(\frac{x}{6}\right) + \frac{x}{2} \sqrt{36 - x^2} + C. \end{aligned}$$

With a little bit of algebra,

$$\frac{x}{2}\sqrt{36-x^2} = 3x\sqrt{1-\frac{x^2}{36}}.$$

So watch out if your solution does not "look" like the "right" one.

Example #4 (Integral involving $a^2 + x^2$)

Find

$$\int \frac{1}{9^2 + x^2} dx.$$

Let

$$x = 9 \tan(t) \quad dx = 9 \sec^2(t) dt$$

for $t \in (-\pi/2, \pi/2)$. We get

$$9^2 + (9 \tan(t))^2 = 9^2 \sec^2(t)$$

so

$$\begin{aligned} \int \frac{1}{9^2 + x^2} dx &= \int \frac{1}{9^2 \sec^2(t)} \cdot 9 \sec^2(t) dt \\ &= \int \frac{1}{9} dt \\ &= \frac{1}{9} t + C. \end{aligned}$$

Now

$$x = 9 \tan(t) \Rightarrow \frac{x}{9} = \tan(t) \Rightarrow t = \arctan\left(\frac{x}{9}\right)$$

so

$$\int \frac{1}{9^2 + x^2} dx = \frac{1}{9} \arctan\left(\frac{x}{9}\right) + C.$$

Example #5 (Integral involving $a^2 + x^2$)

Find

$$\int \frac{1}{\sqrt{16 + x^2}} dx.$$

Let

$$x = 4 \tan(t) \quad dx = 4 \sec^2(t) dt$$

for $t \in (-\pi/2, \pi/2)$. We get

$$4^2 + (4 \tan(t))^2 = 4^2 \sec^2(t)$$

so

$$\begin{aligned} \int \frac{1}{(4^2 + x^2)^{1/2}} dx &= \int \frac{1}{4 \sec(t)} \cdot 4 \sec^2(t) dt \\ &= \int \sec(t) dt \end{aligned}$$

One can show

$$\int \sec(t) dt = \ln |\tan(t) + \sec(t)| + C$$

by multiplying by "1" and letting

$$u = \tan(t) + \sec(t)$$

$$\begin{aligned} du &= (\sec^2(t) + \sec(t) \tan(t)) dt \\ &= (\tan(t) + \sec(t)) \sec(t) dt. \end{aligned}$$

Here is the full computation:

$$\begin{aligned} \int \sec(t) dt &= \int \sec(t) \left(\frac{\tan(t) + \sec(t)}{\tan(t) + \sec(t)} \right) dt \\ &= \int \frac{1}{u} du = \ln |\tan(t) + \sec(t)| + C. \end{aligned}$$

This tells us

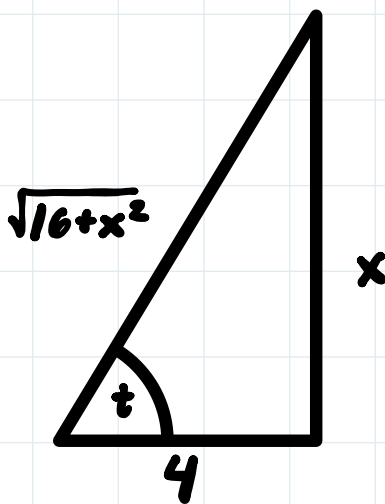
$$\int \frac{1}{(16 + x^2)^{1/2}} dx = \int \sec(t) dt$$

$$= \ln |\tan(t) + \sec(t)| + C.$$

Since

$$x = 4 \tan(t) \Rightarrow \frac{x}{4} = \tan(t)$$

the reference triangle is



and we see

$$\sec(t) = \frac{\sqrt{16 + x^2}}{4}.$$

Thus

$$\int \frac{1}{(16 + x^2)^{1/2}} dx = \ln \left| \frac{x}{4} + \frac{\sqrt{16 + x^2}}{4} \right| + C.$$

We can simplify further

$$\begin{aligned}\int \frac{1}{(16+x^2)^{1/2}} dx &= \ln \left| \frac{x + \sqrt{16+x^2}}{4} \right| + C \\ &= \ln |x + \sqrt{16+x^2}| - \ln(4) + C \\ &= \ln |x + \sqrt{16+x^2}| + C'.\end{aligned}$$

Example #6 (Integral involving $x^2 - a^2$)

Find

$$\int \frac{1}{\sqrt{x^2 - 49}} dx$$

for $x > 7$.

Let

$$x = 7 \sec(t) \quad dx = 7 \sec(t) \tan(t) dt$$

for $t \in (0, \pi/2)$ then

$$7^2 \sec^2(t) - 7^2 = 7^2 \tan^2(t).$$

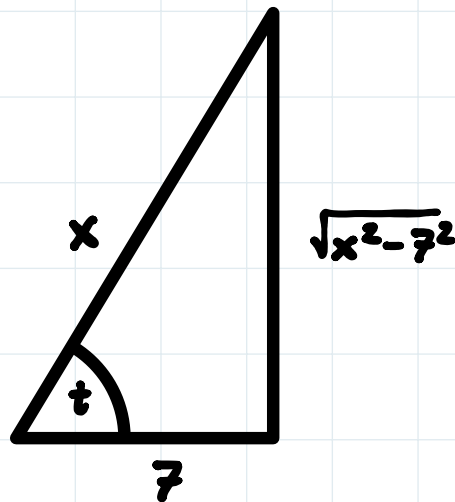
and

$$(x^2 - 49)^{1/2} = 7|\tan(t)| = 7 \tan(t).$$

because of our choice of t .

Now

$$\begin{aligned} \int \frac{1}{(x^2 - 49)^{1/2}} dx &= \int \frac{1}{7 \tan(t)} \cdot 7 \sec(t) \tan(t) dt \\ &= \int \sec(t) dt \\ &= \ln|\sec(t) + \tan(t)| + C \end{aligned}$$



We find

$$x = 7 \sec(t) \Rightarrow \cos(t) = \frac{7}{x} \quad \tan(t) = \frac{\sqrt{x^2 - 7^2}}{7}$$

so

$$\begin{aligned}\int \frac{1}{(x^2 - 49)^{1/2}} dx &= \ln \left| \frac{x}{7} + \frac{\sqrt{x^2 - 7^2}}{7} \right| + C \\ &= \ln \left| \frac{x + \sqrt{x^2 - 7^2}}{7} \right| + C \\ &= \ln |x + \sqrt{x^2 - 7^2}| + C\end{aligned}$$

If you are curious about the $x > 7$ restriction, the function's domain is $(-\infty, -7) \cup (7, \infty)$. The choice of $x > 7$ tells us how the signs of the terms in our reference triangle behave. If we had $x < -7$, we can choose t in $(\pi, 3\pi/2)$ instead of $(0, \pi/2)$.

Example #7

Express

$$x^2 - 2x + 10$$

as

$$u^2 + h$$

where $u = x - k$

Here

$$(x-1)^2 = x^2 - 2x + 1$$

$$\Rightarrow (x-1)^2 + 9 = x^2 - 2x + 1 + 9 = x^2 - 2x + 10$$

Example #8

Find

$$\int \frac{1}{x^2 - 2x + 10} dx.$$

With

$$x^2 - 2x + 10 = (x - 1)^2 + 9$$

let

$$u = x - 1 \quad du = dx.$$

Then

$$\begin{aligned} \int \frac{1}{x^2 - 2x + 10} dx &= \int \frac{1}{(x - 1)^2 + 9} dx \\ &= \int \frac{1}{u^2 + 9} du \end{aligned}$$

Apply another substitution,

$$u = 3 \tan(t) \quad du = 3 \sec^2(t) dt$$

where

$$u^2 + 9 = 9 \tan^2(t) + 9 = 9 \sec^2(t)$$

to get

$$\begin{aligned} \int \frac{1}{u^2 + 9} du &= \int \frac{1}{9 \sec^2(t)} \cdot 3 \sec^2(t) dt \\ &= \frac{1}{3} \int dt \\ &= \frac{1}{3} t + C. \end{aligned}$$

With

$$u = 3 \tan(t) \Rightarrow \frac{u}{3} = \tan(t) \Rightarrow t = \arctan\left(\frac{u}{3}\right).$$

Hence

$$\int \frac{1}{x^2 - 2x + 10} dx = \frac{1}{3} \arctan\left(\frac{x-1}{3}\right) + C.$$

Remark

Here is the summary of the substitutions:

1. $a^2 - x^2$: use $x = a \sin(t)$ and $\cos^2(t) + \sin^2(t) = 1$

2. $a^2 + x^2$: use $x = a \tan(t)$ and $1 + \tan^2(t) = \sec^2(t)$

3. $x^2 - a^2$: use $x = a \sec(t)$ and $1 + \tan^2(t) = \sec^2(t)$

Don't forget,

$$\int \sec(x) \, dx = \ln |\sec(x) + \tan(x)| + C.$$